

# VINUPRAKASH SIVANANTHAM

## Full Stack Engineer

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🐙 [github.com/vinuprakash050](https://github.com/vinuprakash050) 🔗 [portfolio-seven-olive-15.vercel.app](https://portfolio-seven-olive-15.vercel.app)

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### SUMMARY

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Full Stack Engineer with 3.5+ years of experience building enterprise-scale web applications, real-time monitoring systems, and AI-integrated operational platforms using React.js, FastAPI, Python, Node.js, PostgreSQL, and event-driven architectures. Skilled in scalable frontend systems, async backend services, SSE-based communication, Kafka-driven workflows, distributed systems, and enterprise application development.

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### PROFESSIONAL EXPERIENCE

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#### L&T Technology Services Ltd.

Full Stack Developer

12/2022 – Present  
Hyderabad, Telangana

- Developed enterprise-scale full-stack applications using React.js, FastAPI, Python, Node.js, PostgreSQL, SQLAlchemy, and REST APIs.
- Designed and developed 100+ RESTful APIs supporting incident workflows, authentication systems, and real-time monitoring.
- Built scalable async backend services using FastAPI, SQLAlchemy 2.0, AsyncPG, and PostgreSQL with multi-tenant architecture.
- Implemented SSE and WebSocket-based real-time communication systems for live operational dashboards.
- Developed Kafka and Event Mesh based pipelines integrating real-time sensor and camera data with LLM-powered operational workflows.
- Worked on AI-integrated incident processing systems generating contextual recommendations for ICC operators.
- Designed backend architecture using Repository Pattern, Dependency Injection, DTO services, and Domain-Driven Design principles.
- Optimized backend performance through indexing, query optimization, connection pooling, caching, and async concurrency handling.
- Implemented concurrency-safe async workflows using `asyncio.gather` and non-blocking async processing.
- Built reusable React.js dashboard modules and modular frontend systems using Redux and Context API.
- Developed secure authentication and authorization workflows using Azure AD, RBAC, SpiceDB, and middleware systems.
- Independently owned and delivered multiple frontend and backend modules while collaborating directly with clients.

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## TECHNICAL SKILLS

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### Languages

Python, JavaScript (ES6+),  
TypeScript, SQL

### Databases

PostgreSQL, SQLAlchemy 2.0,  
AsyncPG, Alembic, Query  
Optimization

### Authentication

Azure AD, RBAC, SpiceDB,  
CryptoJS, Middleware  
Development

### Frontend

React.js, Redux, Context API,  
HTML5, CSS3, Recharts.js,  
Responsive UI

### Architecture

DDD, Repository Pattern,  
Dependency Injection, DTO  
Architecture, Multi-Tenant  
Systems

### Tools

Git, GitHub, Linux, Loguru,  
Chrome DevTools

### Backend

FastAPI, Node.js, Express.js,  
REST APIs, SSE, Async/Await,  
Pydantic

### Real-Time

Kafka, Event Mesh,  
WebSockets, Pub/Sub Systems,  
Event-Driven Architecture

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## KEY PROJECTS

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### Fusion Worlds.AI

*AI-Powered Incident Management Platform*

- Stack: FastAPI, Python, React.js, PostgreSQL, SQLAlchemy, Kafka, SSE
- Developed enterprise-grade AI-powered incident management platform for ICCO operations.
- Built async backend services using FastAPI, SQLAlchemy 2.0, AsyncPG, and PostgreSQL.
- Implemented Kafka-based publish/subscribe workflows for real-time operational event streaming.
- Developed SSE-based APIs for live incident monitoring and operational dashboards.
- Built backend APIs supporting incident lifecycle management, task assignment, and operational workflows.

### TC4 Smart Building Web Application

*React.js, Redux, Node.js, SQL, REST APIs, Azure AD*

- Developed smart building management platform supporting real-time monitoring and analytics for 300–500 users.
- Built reusable React.js dashboard modules and monitoring interfaces.
- Integrated REST APIs for parking systems, AQI monitoring, and facility analytics.

### Fusion 2.0

*Enterprise Operations & Monitoring Platform*

- Stack: React.js, Node.js, SQL, REST APIs, WebSockets
- Developed incident management systems, schedulers, adaptors, and workflow automation interfaces.
- Built real-time incident tracking systems using WebSockets and operational dashboards.
- Developed backend APIs for incident logging and workflow automation.
- Optimized SQL queries and backend APIs improving system responsiveness.

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## EDUCATION

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### Sona College of Technology

*Bachelor of Engineering (B.E.) – Computer Science and Engineering*  
CGPA: 8.3

2018 – 2022